Name

Mathematics Extension 1 Year 12

Projectile Motion

Book 1

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Syllabus Content

ME1-12-02 operates with 2D and 3D vectors and uses 2D vectors to solve problems involving motion in two dimensions

Projectile motion

- Recognise that the gravitational force on a mass may be regarded as a constant acting in a downwards direction when the motion of the object is restricted to a small region near the Earth's surface
- Model and analyse a projectile's path where the projectile is a point and air resistance is negligible, subject to only acceleration due to gravity, assuming that the projectile is moving close to the Earth's surface
- Represent the motion of a projectile using vectors
- Recognise that the horizontal and vertical components of the motion of a projectile can be represented by horizontal and vertical vectors
- Derive and use the equations of motion of a projectile in vector form by splitting 2D motion into horizontal and vertical components to solve problems on projectiles
- Find the Cartesian equation of the path of a projectile using parametric equations for the horizontal and vertical components of the displacement vector
- Determine features of the flight of a projectile, including time of flight, maximum height, range, instantaneous velocity and impact velocity
- Solve problems relating to the path of a projectile in which the initial velocity and/or angle of projection may be unknown, in a variety of contexts

Projectile Motion and Velocity as a Vector

ⓘ Definition of Projectile Motion

A **projectile** is an object dropped, thrown, struck, or fired into the air near Earth's surface.

We model a projectile as a **single point**.

The **only force** acting on a projectile is **gravity**. We ignore air resistance.

There is **only vertical acceleration** downwards, and **no horizontal acceleration.**

Projectiles	Not Projectiles
A thrown basketball, a kicked football or a golf ball after being hit.	A ball rolling across the ground.
An arrow shot from a bow, a ball shot from a cannon.	Aeroplanes, rockets, or missiles with engines.
Water droplets from a hose, a ball thrown straight up.	A hot air balloon, a person falling with a parachute.

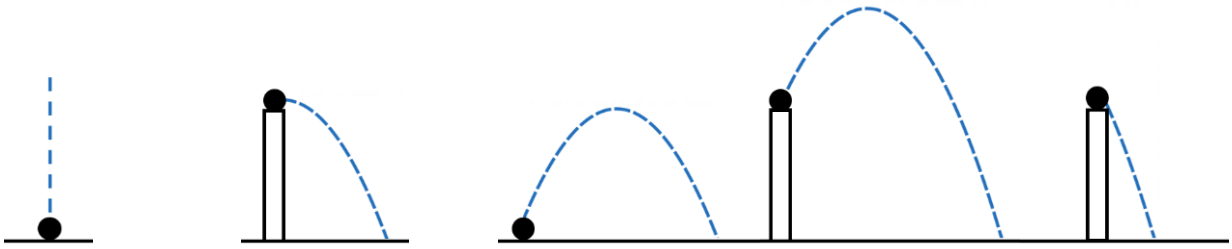
ⓘ Possible Paths of Projectiles

A projectile can be projected vertically, horizontally, or at some angle.

The path is always either a vertical line, a parabola, or section of a parabola:

We model the object as a projectile from when it is projected to when it hits the ground.

Vertical **Horizontal** **Angled upward/downward**



Key Ideas

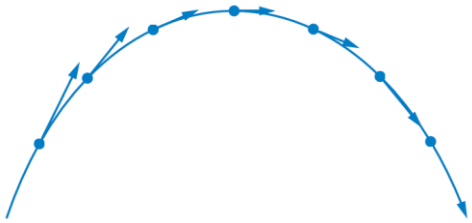
1. For projectile motion, the only force experienced by an object is ……
2. A ball is fired at some angle upwards.
   The vertical acceleration is …… and the horizontal acceleration is ……

ⓘ Projectile Motion as Vectors

In Year 11, you studied motion in a straight line. Here, we extend this to projectile motion in two dimensions.

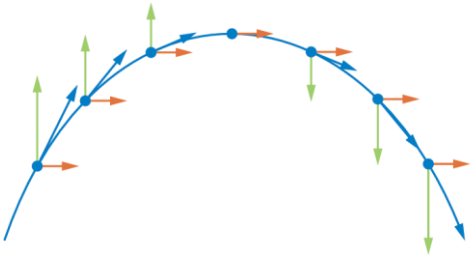
The **displacement** of a projectile traces a **parabolic** curve.

The **velocity vector** at any point is tangent to the path. The direction of travel is therefore $\frac{dy}{dx}$.



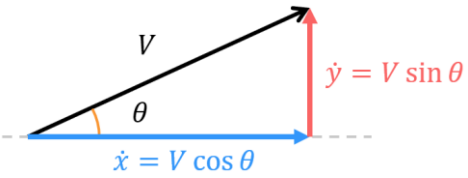
We can **resolve** each velocity vector into **horizontal** and **vertical** components.

The horizontal and vertical components are **independent;** they have no effect on each other.



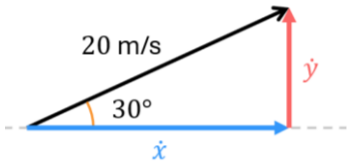
ⓘ Resolving Vectors

For a projectile with speed V and angle of inclination θ,
its **horizontal component** of velocity is $\dot{x} = V\cos\theta$ and
its **vertical component** of velocity is $\dot{y} = V\sin\theta$.
Usually, we treat the right direction and upwards as positive.

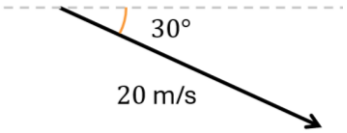


ⓘ Resolve velocity vector into components

a $V = 20$ m/s, $\theta = 30°$ above horizontal



b $V = 20$ m/s, $\theta = 30°$ below horizontal



c $V = 20$ m/s, $\theta = 45°$ above horizontal

d $V = 12$ m/s, $\theta = 60°$ below horizontal

ⓘ Finding Magnitude and Direction

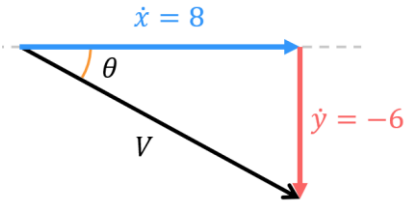
If you are given the horizontal and vertical components and asked for the speed and angle:

$$V = \sqrt{\dot{x}^2 + \dot{y}^2} \qquad \theta = \tan^{-1}\left(\frac{\dot{y}}{\dot{x}}\right)$$

We usually give **negative angles** as **'below the horizontal'**

ⓘ Calculate vector magnitude and direction from components

a A projectile has velocity components $\dot{x} = 8$ m/s and $\dot{y} = -6$ m/s.



b A projectile has velocity components $\dot{x} = 3$ m/s and $\dot{y} = 3\sqrt{3}$ m/s

Deriving the Six Equations of Motion

ⓘ The Six Equations of Motion

The three equations of motion in one dimension describe **displacement**, **velocity**, and **acceleration** with respect to time t, related by differentiation and integration.

$x(t)$: displacement $\quad$ $\dot{x}(t)$: velocity $\quad$ $\ddot{x}$: acceleration

In projectile motion, there are six equations of motion:

three for horizontal motion and three for vertical motion.

Every projectile motion problem starts from two facts:

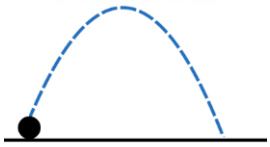
- $\ddot{x}(t) = 0$ No horizontal force acts, so there is no horizontal acceleration.
- $\ddot{y}(t) = -g$ The only vertical force is the constant force of gravity.

The remaining four equations follow by integrating these twice.

ⓘ Initial Conditions for Projection from the ground

During integration, we need **initial conditions** for **velocity** and **displacement** to find the particular solutions of the velocity and displacement equations.

When a projectile is fired from the ground, we set the projection point as the **origin:**



$x(0) = 0$

$\dot{x}(0) = V\cos\theta$

$\ddot{x}(0) = 0$

$y(0) = 0$

$\dot{y}(0) = V\sin\theta$

$\ddot{y}(0) = -g$

Review

☐ Integrate polynomials

a $\int 0\,dt$

b $\int -10\,dt$

c $\int -10t + 20\,dt$

☐ Find constants of integration

a Given $y = 2t + C$ and $y(0) = 20\sqrt{3}$, find C.

b Given $y = -10t^2 + 5t + C$ and $y(0) = 0$, find C.

Guided Practice

A ball is thrown from the origin with speed 40 m/s at an angle of 30° to the horizontal.

Taking $g = 10$ m/s², derive the six equations of motion.

Acceleration: $\ddot{x} = 0$

Velocity:

$\dot{x} = 20\sqrt{3}$

Displacement:

$x = 20\sqrt{3}t$

Acceleration: $\ddot{y} = -10$

Velocity:

$\dot{y} = -10t + 20$

Displacement:

$y = -5t^2 + 20t$

A projectile is fired from the origin at 20 m/s at 60° to the horizontal, with $g = 9.8$ m/s².

$\dot{x} = 10$

$x = 10t$

$\dot{y} = -9.8t + 10\sqrt{3}$

$y = -4.9t^2 + 10\sqrt{3}t$

ⓘ General form of the Equations

By deriving the six equations using the general initial velocity of V m/s at θ above horizontal:

	Horizontal	Vertical
Acceleration	$\ddot{x} = 0$	$\ddot{y} = -g$
Velocity	$\dot{x} = V\cos\theta$	$\dot{y} = -gt + V\sin\theta$
Displacement	$x = Vt\cos\theta$	$y = -\frac{1}{2}gt^2 + Vt\sin\theta$

You are expected to derive the six equations, unless given to you in a question.

You may not memorise them in an exam as there will be marks for the derivation.

In the next few sections of the booklet, we will skip the derivation to save time.

ⓘ Vector form of the equations

The six equations above give the scalar components for each direction.

We can combine them to vector form for acceleration, velocity, and displacement:

$$\underset{\sim}{a} = 0\boldsymbol{i} - g\boldsymbol{j} = \begin{pmatrix} 0 \\ -g \end{pmatrix}$$

$$\underset{\sim}{v} = (V\cos\theta)\boldsymbol{i} + (-gt + V\sin\theta)\boldsymbol{j}$$

$$\underset{\sim}{r} = (Vt\cos\theta)\boldsymbol{i} + \left(-\frac{1}{2}gt^2 + Vt\sin\theta\right)\boldsymbol{j}$$

Displacement can also be $\underset{\sim}{d}$ or $\underset{\sim}{s}$.

ⓘ **Rewrite** scalar components as vectors

$\dot{x} = 10$

$\dot{y} = -10t + 10\sqrt{3}$

$\underset{\sim}{v} = 10\,\boldsymbol{i} + (-10t + 10\sqrt{3})\,\boldsymbol{j}$

$x = 10t$

$y = -5t^2 + 10\sqrt{3}\,t$

$\underset{\sim}{r} = 10t\,\boldsymbol{i} + (-5t^2 + 10\sqrt{3}\,t)\,\boldsymbol{j}$

Write these components as a vector.

a $\dot{x} = 20,\ \dot{y} = -10t + 20$

b $x = 20t,\ y = -5t^2 + 20t$

c $\ddot{x} = 0,\ \ddot{y} = -g$

d $x = 10t,\ y = -5t^2 + 10t + 5$

$\underset{\sim}{r} = \left(\quad\quad\right)$

ⓘ Finding Equations of Motion using Vector Form

You can find the vector form of the equations by finding the six equations and integrating.

You can also find the equations of motion by integrating the whole vector.

ⓒ **Compare** deriving six equations of motion using vector form

A projectile is fired from the origin at 20 m/s at 60° to the horizontal. Take $g = 10$ m/s²

Component derivation

Acceleration:

$$\ddot{x} = 0 \qquad \ddot{y} = -10$$

Velocity:

$$\dot{x} = \int 0 = C_1 \qquad \dot{y} = \int -10\,dt$$

$$\dot{x}(0) = 20\cos 60° \qquad = -10t + C_3$$

$$= 10 \qquad \dot{y}(0) = 20\sin 60°$$

$$10 = C_1 \qquad = 10\sqrt{3}$$

$$10\sqrt{3} = -10(0) + C_3$$

$$10\sqrt{3} = C_3$$

$$\therefore \dot{x} = 10 \qquad \therefore \dot{y} = -10t + 10\sqrt{3}$$

Displacement:

$$x = \int 10\,dt \qquad y = \int -10t + 10\sqrt{3}\,dt$$

$$= 10t + C_2 \qquad = -5t^2 + 10\sqrt{3}t + C_4$$

$$x(0) = 0 \qquad y(0) = 0$$

$$0 = 10(0) + C_2 \qquad 0 = -5(0)^2 10\sqrt{3}(0) + C_4$$

$$C_2 = 0 \qquad C_4 = 0$$

$$\therefore x = 10t \qquad \therefore y = -5t^2 + 10\sqrt{3}t$$

Vector form:

$$\underset{\sim}{a} = \begin{pmatrix} 0 \\ -10 \end{pmatrix}$$

$$\underset{\sim}{v} = \begin{pmatrix} 10 \\ -10t + 10\sqrt{3} \end{pmatrix}$$

$$\underset{\sim}{r} = \begin{pmatrix} 10t \\ -5t^2 + 10\sqrt{3}\,t \end{pmatrix}$$

Vector derivation

Acceleration:

$$\underset{\sim}{a} = \begin{pmatrix} 0 \\ -10 \end{pmatrix}$$

Velocity:

$$\underset{\sim}{v} = \int \begin{pmatrix} 0 \\ -10 \end{pmatrix} dt$$

$$= \begin{pmatrix} C_1 \\ -10t + C_2 \end{pmatrix}$$

$$\text{using } \underset{\sim}{v}(0) = \begin{pmatrix} 10 \\ 10\sqrt{3} \end{pmatrix} = \begin{pmatrix} C_1 \\ -10(0) + C_2 \end{pmatrix}$$

$$\begin{pmatrix} 10 \\ 10\sqrt{3} \end{pmatrix} = \begin{pmatrix} C_1 \\ C_2 \end{pmatrix}$$

$$\underset{\sim}{v} = \begin{pmatrix} 10 \\ -10t + 10\sqrt{3} \end{pmatrix}$$

Displacement:

$$\underset{\sim}{r} = \int \begin{pmatrix} 10 \\ -10t + 10\sqrt{3} \end{pmatrix} dt$$

$$= \begin{pmatrix} 10t + C_3 \\ -5t^2 + 10\sqrt{3}t + C_4 \end{pmatrix}$$

$$\underset{\sim}{r}(0) = \begin{pmatrix} 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 10(0) + C_3 \\ -5(0)^2 + 10\sqrt{3}(0) + C_4 \end{pmatrix}$$

$$\begin{pmatrix} C_3 \\ C_4 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\underset{\sim}{r} = \begin{pmatrix} 10t \\ -5t^2 + 10\sqrt{3}t \end{pmatrix}$$

Take $g = 10\ \text{ms}^{-2}$

1 A particle is projected from the origin with speed 20 m/s at 60° to the horizontal. By integrating, find the velocity and displacement, and write the acceleration, velocity, and displacement in vector form.

Shown here is the derivation using $\boldsymbol{i}$ and $\boldsymbol{j}$ notation

$$\boldsymbol{a} = 0\boldsymbol{i} - 10\boldsymbol{j}$$

$$\boldsymbol{v} = C_1\boldsymbol{i} + (-10t + C_2)\boldsymbol{j}$$

$$\boldsymbol{v}(0) = 20\cos 60°\boldsymbol{i} + 20\sin 60°\boldsymbol{j}$$

$$\boldsymbol{v}(0) = 10\boldsymbol{i} + 10\sqrt{3}\boldsymbol{j}$$

$$C_1 = 10, \qquad C_2 = 10\sqrt{3}$$

$$\boxed{\boldsymbol{v} = 10\boldsymbol{i} + (10\sqrt{3} - 10t)\boldsymbol{j}}$$

$$\boldsymbol{r} = (10t + C_3)\boldsymbol{i} + \left(10\sqrt{3}t - 5t^2 + C_4\right)\boldsymbol{j}$$

$$\boldsymbol{r}(0) = 0\boldsymbol{i} + 0\boldsymbol{j}$$

$$C_3 = 0, \qquad C_4 = 0$$

$$\boxed{\boldsymbol{r} = 10t\boldsymbol{i} + \left(10\sqrt{3}t - 5t^2\right)\boldsymbol{j}}$$

$$\boxed{\boldsymbol{a} = -10\boldsymbol{j}}$$

2 A ball is thrown from the origin with speed 20 m/s at 30°.
Derive the six equations of motion and write $\underset{\sim}{v}$ in vector form.

$$u_x = 20\cos 30° = 10\sqrt{3}$$

$$u_y = 20\sin 30° = 10$$

$$a_x = 0, \qquad a_y = -10$$

$$\frac{dv_x}{dt} = 0$$

$$v_x = C_1$$

$$v_x(0) = 10\sqrt{3}$$

$$C_1 = 10\sqrt{3}$$

$$\boxed{v_x = 10\sqrt{3}}$$

$$\frac{dv_y}{dt} = -10$$

$$v_y = -10t + C_2$$

$$v_y(0) = 10$$

$$C_2 = 10$$

$$\boxed{v_y = 10 - 10t}$$

$$\frac{dx}{dt} = v_x = 10\sqrt{3}$$

$$x = 10\sqrt{3}t + C_3$$

$$x(0) = 0$$

$$C_3 = 0$$

$$\boxed{x = 10\sqrt{3}t}$$

$$\frac{dy}{dt} = v_y = 10 - 10t$$

$$y = 10t - 5t^2 + C_4$$

$$y(0) = 0$$

$$C_4 = 0$$

$$\boxed{y = 10t - 5t^2}$$

$$\boxed{a_x = 0}$$

$$\boxed{a_y = -10}$$

$$\boxed{v_x = 10\sqrt{3}}$$

$$\boxed{v_y = 10 - 10t}$$

$$\boxed{x = 10\sqrt{3}t}$$

$$\boxed{y = 10t - 5t^2}$$

$$\boxed{\boldsymbol{v} = 10\sqrt{3}\boldsymbol{i} + (10 - 10t)\boldsymbol{j}}$$

3 A ball is thrown from the origin with speed 40 m/s at $\tan^{-1}\left(\frac{3}{4}\right)$.
Derive the six equations and write $\underset{\sim}{r}$ in vector form.

$$\theta = \tan^{-1}\left(\frac{3}{4}\right)$$
$$\sin\theta = \frac{3}{5}, \qquad \cos\theta = \frac{4}{5}$$
$$u_x = 40\cos\theta = 40 \cdot \frac{4}{5} = 32$$
$$u_y = 40\sin\theta = 40 \cdot \frac{3}{5} = 24$$
$$a_x = 0, \qquad a_y = -10$$
$$\frac{dv_x}{dt} = 0$$
$$v_x = C_1$$
$$v_x(0) = 32$$
$$C_1 = 32$$
$$\boxed{v_x = 32}$$
$$\frac{dv_y}{dt} = -10$$
$$v_y = -10t + C_2$$
$$v_y(0) = 24$$
$$C_2 = 24$$
$$\boxed{v_y = 24 - 10t}$$
$$\frac{dx}{dt} = v_x = 32$$
$$x = 32t + C_3$$
$$x(0) = 0$$
$$C_3 = 0$$
$$\boxed{x = 32t}$$
$$\frac{dy}{dt} = v_y = 24 - 10t$$
$$y = 24t - 5t^2 + C_4$$
$$y(0) = 0$$
$$C_4 = 0$$
$$\boxed{y = 24t - 5t^2}$$

$$\boxed{a_x = 0}$$
$$\boxed{a_y = -10}$$
$$\boxed{v_x = 32}$$
$$\boxed{v_y = 24 - 10t}$$
$$\boxed{x = 32t}$$
$$\boxed{y = 24t - 5t^2}$$
$$\boxed{\boldsymbol{r} = 32t\boldsymbol{i} + (24t - 5t^2)\boldsymbol{j}}$$

4 A particle is projected from the origin with speed V at angle θ. By integrating, show that $\underset{\sim}{v} = (V\cos\theta)\,\boldsymbol{i} + (-10t + V\sin\theta)\,\boldsymbol{j}$ and $\underset{\sim}{r} = (Vt\cos\theta)\,\boldsymbol{i} + (-5t^2 + Vt\sin\theta)\boldsymbol{j}$. Take $g = 10$ m/s².

$$\boldsymbol{a} = 0\boldsymbol{i} - 10\boldsymbol{j}$$

$$\boldsymbol{v} = \int (0\boldsymbol{i} - 10\boldsymbol{j})\, dt$$
$$\boldsymbol{v} = C_1\boldsymbol{i} + (-10t + C_2)\boldsymbol{j}$$
$$\boldsymbol{v}(0) = V\cos\theta\boldsymbol{i} + V\sin\theta\boldsymbol{j}$$
$$C_1 = V\cos\theta$$
$$C_2 = V\sin\theta$$
$$\boxed{\boldsymbol{v} = (V\cos\theta)\boldsymbol{i} + (-10t + V\sin\theta)\boldsymbol{j}}$$

$$\boldsymbol{r} = \int [(V\cos\theta)\boldsymbol{i} + (-10t + V\sin\theta)\boldsymbol{j}]\, dt$$
$$\boldsymbol{r} = (Vt\cos\theta + C_3)\boldsymbol{i} + (-5t^2 + Vt\sin\theta + C_4)\boldsymbol{j}$$
$$\boldsymbol{r}(0) = 0\boldsymbol{i} + 0\boldsymbol{j}$$
$$C_3 = 0$$
$$C_4 = 0$$
$$\boxed{\boldsymbol{r} = (Vt\cos\theta)\boldsymbol{i} + (-5t^2 + Vt\sin\theta)\boldsymbol{j}}$$

Using the Equations of Motion

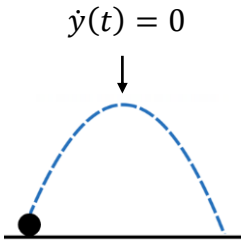
ⓘ Maximum Height is where $\dot{y} = 0$

A projectile rises, reaches a maximum height, then falls.

Vertical velocity is **positive while rising** and **negative while falling**,

At the maximum height, where the motion changes direction, the vertical velocity is zero.

To find the maximum height, set $\dot{y}(t) = 0$ to find the time at which the projectile reaches maximum height, then substitute into $y(t)$.

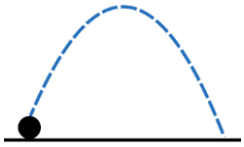


ⓘ Projectile lands when $y = 0$

For a projectile launched from ground level, it lands when it returns to ground level, $y(t) = 0$.

Setting $y(t) = 0$ gives two solutions: $t = 0$ (launch) and a $t =$ positive value (landing).

The time of landing is also called the **time of flight**.

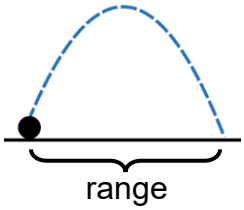


$t = 0$ $t =$ positive value (time of flight)

ⓘ Range

The range is the total horizontal distance travelled.

To find the range, first find the time of flight from $y = 0$, then substitute into $x(t)$.



ⓘ Summary of Projectile Motion Quantities

Quantity		How to find
Time for maximum height	$t_{y\,\max}$	Solve $\dot{y}(t) = 0$
Maximum height	$y_{\max}$	Substitute $t_{y\,\max}$ into $y(t)$
Time of flight (when it lands)	t_{flight}	Solve $y(t) = 0$
Range	$x_{\max}$	Substitute t_{flight} into $x(t)$

ⓘ **Identify** the method for finding a projectile motion quantity

Decide which equation to use to find each of these quantities.

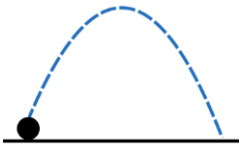
Some may require you to find a specific time first.

a	Find horizontal displacement at $t = 3$	**b**	Find the time to reach maximum height
c	Find the time of flight	**d**	Find the maximum height
e	Find the horizontal velocity at $t = 3$	**f**	Find the time when the projectile is 15 m above the ground.
g	Find the range	**h**	Find the speed when the projectile lands
i	Find the angle the projectile lands	**j**	Find when the horizontal displacement is 6 m from the origin

Example Solve projectile motion problems projected from the ground

A ball is hit from the ground with speed 20 m/s at 45° to the horizontal. Take $g = 10$ m/s². Find

a the six equations of motion
b the time to reach maximum height
c the maximum height
d the total time of flight
e the range
f the impact velocity



a)

full derivation skipped

$\ddot{x} = 0$ $\qquad$ $\ddot{y} = -10$

$\dot{x} = 20\cos 45°$ $\qquad$ $\dot{y} = -10t + 20\sin 45°$

$= 10\sqrt{2}$ $\qquad$ $= -10t + 10\sqrt{2}$

$x = 10\sqrt{2}t$ $\qquad$ $y = -5t^2 + 10\sqrt{2}t$

b)

Let $\dot{y}(t) = 0$

$0 = -10t + 10\sqrt{2}$

$10t = 10\sqrt{2}$

$t = \sqrt{2}$ s

c)

$y(\sqrt{2}) = -5(\sqrt{2})^2 + 10\sqrt{2}(\sqrt{2})$

$= 10$ m

d)

Let $y(t) = 0$

$0 = -5t^2 + 10\sqrt{2}t$

$0 = t(-5t + 10\sqrt{2})$

$t = 0$ or $-5t + 10\sqrt{2} = 0$

$\therefore t = 0$ or $2\sqrt{2}$ s

e)

$x(2\sqrt{2}) = 10\sqrt{2}(2\sqrt{2})$

$= 40$ m

f)

$\dot{y}(2\sqrt{2}) = -10(2\sqrt{2}) + 10\sqrt{2} = -10\sqrt{2}$

$\dot{x}(2\sqrt{2}) = 10\sqrt{2}$

$V = \sqrt{(-10\sqrt{2})^2 + (10\sqrt{2})^2} = 20$

$\theta = \tan^{-1}\left(\frac{-10\sqrt{2}}{10\sqrt{2}}\right) = -45°$

20 m/s at 45° below the horizontal

Symmetry of the Projectile's Path

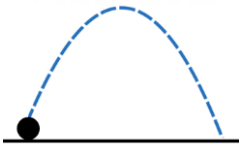
For projection from the ground, the path of a projectile is a symmetric parabola.
As seen from the example, the time of flight is exactly double the time for max height.
The impact velocity is the same as the starting velocity, except the $\dot{y}$ component is negative.

Generally, you may use properties of a parabola to shortcut projectile motion problems.

Guided Practice

A ball is hit from ground level at 30 m/s at $30°$ to the horizontal, $g = 10$ m/s². Find:

a the six equations of motion
b the time to reach maximum height
c the maximum height
d the total time of flight
e the range
f the impact velocity



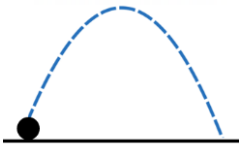
a)

$\ddot{x} = 0$ $\qquad$ $\ddot{y} = -10$
$\dot{x} = 15\sqrt{3}$ $\qquad$ $\dot{y} = -10t + 15$
$x = 15\sqrt{3}t$ $\qquad$ $y = -5t^2 + 15t$

b) $t_{y\,\max} = 1.5$ s
c) $y_{\max} = 11.25$ m
d) $t_{\text{tof}} = 3$ s
e) $x_{\max} = 77.9$ m
f) 30 m/s at 30° below horizontal

A ball is thrown from the ground at 50 m/s at 45°, $g = 10$.

Find the velocity when the ball is at height 40 m to 2 d.p.



41.2 m/s at either 30.96° above horizontal or 30.96° below horizontal

Let $g = 10$ m/s² unless otherwise stated.

1 A particle is projected from the origin with a speed of $30\sqrt{2}$ m/s at an angle of 45° to the horizontal.

a Starting with $\ddot{x} = 0$ and $\ddot{y} = -10$, show that $\dot{x} = 30$ and $\dot{y} = -10t + 30$.

$\ddot{x} = 0 \rightarrow \dot{x} = C_1;$

at $t = 0: \dot{x} = 30\sqrt{2}\cos 45° = 30 \rightarrow C_1 = 30$

$\therefore \dot{x} = 30$

$\ddot{y} = -10 \rightarrow \dot{y} = -10t + C_2;$

at $t = 0: \dot{y} = 30\sqrt{2}\sin 45° = 30 \rightarrow C_2 = 30$

$\therefore \dot{y} = -10t + 30$

b Hence find x and y in terms of t.

$x = 30t$

$y = -5t^2 + 30t$

c Put $y = 0$ to find when the particle returns to the x-axis.

$y = 0: 0 = t(-5t + 30)$

$t = 6$ s

d Hence find the horizontal distance travelled.

$x(6) = 180$ m

e Put $\dot{y} = 0$ to find when the particle reaches its greatest height.

$\dot{y} = 0: -10t + 30 = 0$

$t = 3$ s

f Find the greatest height.

$y(3) = -5(9) + 30(3) = 45$ m

2 A particle is projected from horizontal ground with a speed of 40 m/s at an angle of 30° to the horizontal.

a Starting with $\ddot{x} = 0$ and $\ddot{y} = -10$, show that $\dot{x} = 20\sqrt{3}$ and $\dot{y} = -10t + 20$.

$\dot{x} = 40\cos 30° = 20\sqrt{3}$

$\dot{y} = -10t + 40\sin 30° = -10t + 20$

b Hence find x and y in terms of t.

$x = 20\sqrt{3}t$

$y = -5t^2 + 20t$

c Find:

i when the particle returns to the ground,

ii the horizontal distance travelled,

iii the greatest height reached.

i) $y = 0: t(-5t + 20) = 0$

$t = 4$ s

ii) $x(4) = 80\sqrt{3}$ m

iii) $\dot{y} = 0: t = 2$

$y(2) = -20 + 40 = 20$ m

3 A particle is projected from the origin. Its initial velocity vector is $8\boldsymbol{i} + 6\boldsymbol{j}$.

a Express its velocity at time t in the form $\underset{\sim}{v} = \dot{x}\boldsymbol{i} + \dot{y}\boldsymbol{j}$.

$$\underset{\sim}{v} = 8\boldsymbol{i} + (6 - 10t)\boldsymbol{j}$$

b Express its displacement at time t in the form $\underset{\sim}{r} = x\boldsymbol{i} + y\boldsymbol{j}$.

$$\underset{\sim}{r} = 8t\boldsymbol{i} + (6t - 5t^2)\boldsymbol{j}$$

c Find:

i the initial speed,

ii the position vector after 2 seconds,

iii the position vector when it reaches its greatest height.

$$\text{i) Speed} = \sqrt{64 + 36} = 10 \text{ m/s}$$

$$\text{ii) } \underset{\sim}{r}(2) = 16\boldsymbol{i} + (12 - 20)\boldsymbol{j} = 16\boldsymbol{i} - 8\boldsymbol{j}$$

$$\text{iii) } \dot{y} = 0 : t = 0.6;$$

$$\underset{\sim}{r}(0.6) = 4.8i + (3.6 - 1.8)j = 4.8i + 1.8j$$

4 A particle is projected from the origin. Its horizontal and vertical displacement after t seconds are $x = 40t$ and $y = -5t^2 + 25t$.

a Find the initial values of $\dot{x}$ and $\dot{y}$.

$$\text{At } t = 0 : \dot{x} = 40, \dot{y} = 25$$

b Hence show the initial velocity is $5\sqrt{89}$ at an angle of approximately 32° to the horizontal.

$$\text{Speed} = \sqrt{1600 + 625} = 5\sqrt{89}$$

$$\theta = \tan^{-1}\left(\frac{25}{40}\right) = \tan^{-1}\left(\frac{5}{8}\right) \approx 32°$$

5 A particle is projected from the origin with a speed of 20 m/s at an angle of 60° to the horizontal.

a Starting with $\ddot{x} = 0$ and $\ddot{y} = -10$, show that $\dot{x} = 10$ and $\dot{y} = -10t + 10\sqrt{3}$.

$$\dot{x} = 20\cos 60° = 10$$

$$\dot{y} = -10t + 20\sin 60° = -10t + 10\sqrt{3}$$

b Hence find x and y in terms of t.

$$x = 10t,\ y = -5t^2 + 10\sqrt{3}\,t$$

c Use Pythagoras' theorem to find, correct to one decimal place:

i the distance of the particle from the origin after one second,

ii the speed of the particle after one second.

$$\text{i) } At\ t = 1 : x = 10, y = 10\sqrt{3} - 5$$

$$\text{Distance} = \sqrt{10^2 + \left(10\sqrt{3} - 5\right)^2} \approx 15.9 \text{ m}$$

$$\text{ii) } \dot{x} = 10, \dot{y} = 10\sqrt{3} - 10$$

$$\text{speed} = \sqrt{100 + 100\left(\sqrt{3} - 1\right)^2} \approx 12.4 \text{ m/s}$$

6 A particle is projected from ground level with a speed of 60 m/s at an angle of $\tan^{-1}\left(\frac{4}{3}\right)$ to the horizontal.

a Show that $\dot{x} = 36$ and $\dot{y} = -10t + 48$.

$\tan\theta = \frac{4}{3}$

$\sin\theta = \frac{4}{5}, \cos\theta = \frac{3}{5}$

$\dot{x} = 60\left(\frac{3}{5}\right) = 36;$

$\dot{y} = -10t + 60\left(\frac{4}{5}\right) = -10t + 48$

b Find x and y as functions of t.

$x = 36t$

$y = -5t^2 + 48t$

c Find, correct to one decimal place, the distance of the particle from the point of projection after 3 seconds.

At $t = 3: x = 108$

$y = -45 + 144 = 99$

Distance $= \sqrt{108^2 + 99^2} \approx 146.5$ m

d Show that the velocity after 3 seconds is $18\sqrt{5}$ m/s at an angle of $\tan^{-1}\left(\frac{1}{2}\right)$ above the horizontal.

$\dot{x}(3) = 36, \dot{y}(3) = -30 + 48 = 18$

Speed $= \sqrt{36^2 + 18^2} = 18\sqrt{5}$

$\theta = \tan^{-1}\left(\frac{18}{36}\right) = \tan^{-1}\left(\frac{1}{2}\right)$ above horizontal

7 A particle is projected at an angle of 45° and a speed of 15 m s⁻¹.

a Derive the equation of the displacement vector.

b Find the time taken to reach the ground.

a) $\ddot{x} = 0, \ddot{y} = -10;$

$\dot{x} = 15\cos 45° = \frac{15\sqrt{2}}{2}, \dot{y} = -10t + \frac{15\sqrt{2}}{2}$

$s = \left(\frac{15\sqrt{2}}{2}t\right)\boldsymbol{i} + \left(-5t^2 + \frac{15\sqrt{2}}{2}t\right)\boldsymbol{j}$

b) $y = 0: t\left(\frac{15\sqrt{2}}{2} - 5t\right) = 0$

$t = \frac{3\sqrt{2}}{2}$ s

8 A bullet is fired at an angle of 60° and a speed of 120 m s⁻¹. Find:

a the exact time taken to reach maximum height,

b the bullet's maximum height.

a) $\dot{y} = -10t + 120\sin 60° = -10t + 60\sqrt{3}$

Let $\dot{y} = 0$

$t = 6\sqrt{3}$ s

b) $y(6\sqrt{3}) = 540$ m

9 An object is projected with horizontal velocity 8 m s⁻¹ and vertical velocity 5 m s⁻¹. Find the range. Use g = 10 m s⁻².

$x = 8t$

$y = -5t^2 + 5t$

Let $y = 0$

$-5t^2 + 5t = 0$

$5t(-t + 1) = 0$

$\therefore t_{\text{flight}} = 1$

Range = $x(1) = 8$ m

10 A particle is projected at 10 m s⁻¹. The horizontal component of velocity is 6 m s⁻¹. Find:

a the vertical component of velocity,

b the angle of projection,

c the maximum height (use g = 10 m s⁻²),

d the speed after:

i 2 s

ii 3.1 s

a) $\dot{y} = \sqrt{100 - 36} = 8\ m\ s^{-1}$

b) $\theta = \tan^{-1}\left(\frac{8}{6}\right) = \tan^{-1}\left(\frac{4}{3}\right) \approx 53°$

c) max height = $\frac{8^2}{2 \times 10} = 3.2$ m

d)

i) $\dot{x} = 6, \dot{y}(2) = 8 - 20 = -12;$

speed $= \sqrt{36 + 144} = 6\sqrt{5} \approx 13.4$ m s^{-1}

ii) $\dot{y}(3.1) = 8 - 31 = -23;$

speed $= \sqrt{565} \approx 23.8$ m s^{-1}

11 Pham throws a ball at a velocity of 8.7 m s⁻¹ at an angle of 55°.
Use g = 9.8 m s⁻² and neglect air resistance. Find:

a the maximum height reached (to 2 d.p.),

b how far from Pham it lands (to 1 d.p.).

$a)\ \dot{x} = 8.7\cos 55°, \dot{y} = -9.8t + 8.7\sin 55°$

$t_{max} = \dfrac{8.7\sin 55°}{9.8} \approx 0.7273\text{ s}$

$y_{max} = -4.9(0.7273)^2 + 8.7\sin 55°(0.7273) \approx 2.59\text{ m}$

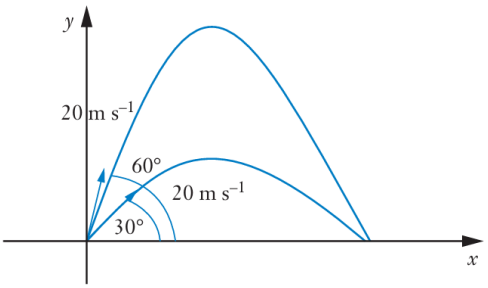
b) $t_{flight} = 2(0.7273) \approx 1.455\text{ s}$

$x = 8.7\cos 55°(1.455) \approx 7.3\text{ m}$

12 A stone is thrown at 20 m s⁻¹ at 60°. At the same time and place, a second stone is thrown at the same speed but at 30°. Use g = 10 m s⁻².

a Which stone finishes its flight first and by how long?

b Show that the 2 stones land at the same place.



a)

Stone 1 (60°)

$x = 10t, y = 10\sqrt{3}t - 5t^2$

Stone 2 (30°)

$x = 10\sqrt{3}t, y = 10t - 5t^2$

$t_{flight}(60°): t(10\sqrt{3} - 5t) = 0 \rightarrow t = 2\sqrt{3}\text{ s}$

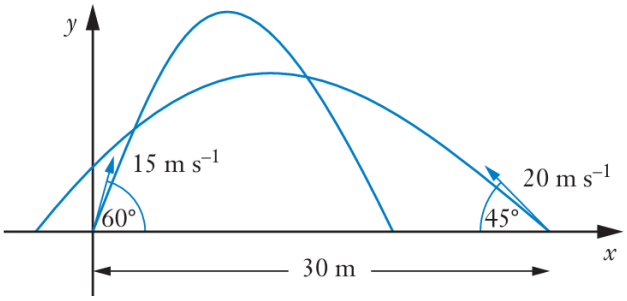
$t_{flight}(30°): t(10 - 5t) = 0 \rightarrow t = 2\text{ s}$

Stone at 30° lands first, by $2\sqrt{3} - 2 \approx 1.5\text{ s}$

b) $\text{Range}(60°) = \dfrac{400\sin 120°}{10} = 20\sqrt{3}$

$\text{Range}(30°) = \dfrac{400\sin 60°}{10} = 20\sqrt{3}$

13 A ball is thrown at 15 m s⁻¹ at 60°. At the same time another ball is thrown from a point 30 m away, back towards the first, at 45° and 20 m s⁻¹. How far apart do they land? Use g = 10 m s⁻².



Ball 1 (from x = 0, rightward, θ = 60°, V = 15):

$\text{Range} = \dfrac{225\ sin\ 120°}{10} = \dfrac{225\sqrt{3}}{20} \approx 19.49\text{ m};$

lands at $x \approx 19.49$

Ball 2 (from x = 30, leftward, θ = 45°, V = 20):

$\text{Range} = \dfrac{400\ sin\ 90°}{10} = 40\text{ m};$

lands at $x = 30 - 40 = -10\text{ m}$

Distance between landing spots = $19.49 - (-10) \approx 29.5\text{ m}$

Launching from a Platform

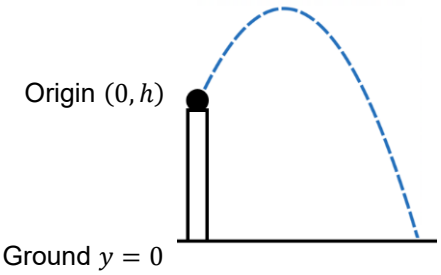
ⓘ Projection from a Platform

When a projectile launches from height h above the ground, there are two approaches:

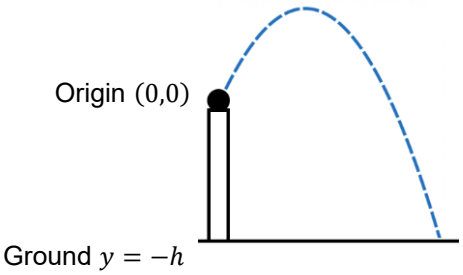
Set the initial vertical displacement to h; the equation of vertical displacement becomes a more complex quadratic.

Keep the origin at the launch point and set ground level to $-h$; the initial conditions stay the same, but the landing condition becomes $y(t) = -h$

Option 1:



Option 2:



Mr Ding recommends option 2.

So, when a projectile is fired from a **platform, treat the platform as the origin**.

All our initial conditions stay the same.

$$x(0) = 0 \qquad y(0) = 0$$
$$\dot{x}(0) = V\cos\theta \qquad \dot{y}(0) = V\sin\theta$$
$$\ddot{x}(0) = 0 \qquad \ddot{y}(0) = -g$$

So, our equations of motion stay the same.

	Horizontal	Vertical
Acceleration	$\ddot{x} = 0$	$\ddot{y} = -g$
Velocity	$\dot{x} = V\cos\theta$	$\dot{y} = -gt + V\sin\theta$
Displacement	$x = Vt\cos\theta$	$y = -\frac{1}{2}gt^2 + Vt\sin\theta$

⚠ When using the equations, treat the platform as $y = 0$

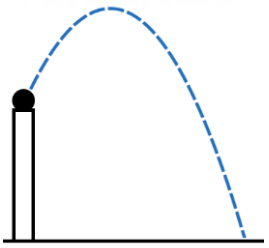
⚠ But when answering a question, add the height of the platform in your final answer.
E.g. if you find max height is $y = 3$, and your platform is 20 m high, answer as 23 metres

For all questions in this section take $g = 10$ m/s² unless otherwise stated.

Example Solve projectile motion problems projected from a platform

A ball is thrown from the top of a 25 m cliff with speed 40 m/s at 30° above the horizontal. Taking the top of the cliff as the origin:

a Find the six equations of motion.
b Find the time the ball takes to hit the ground at the base of the cliff.
c Find the horizontal distance from the base of the cliff to where the ball lands to 2 d.p.
d Find the impact speed and angle to 2 d.p.
e Find the maximum height of the ball.



a) full derivation skipped

$$\underset{\sim}{a} = \begin{pmatrix} 0 \\ -10 \end{pmatrix}$$

$$\underset{\sim}{v} = \begin{pmatrix} 40\cos 30° \\ -10t + 40\sin 30° \end{pmatrix}$$

$$= \begin{pmatrix} 20\sqrt{3} \\ -10t + 20 \end{pmatrix}$$

$$\underset{\sim}{r} = \begin{pmatrix} 20\sqrt{3}t \\ -5t^2 + 20t \end{pmatrix}$$

b)

Let $y(t) = -25$

$$-25 = -5t^2 + 20t$$

$$5t^2 - 20t - 25 = 0$$

$$t^2 - 4t - 5 = 0$$

$$(t-5)(t+1) = 0$$

$$t = 5 \ (t \geq 0)$$

c)

$$x(5) = 20\sqrt{3}(5) = 100\sqrt{3}$$

$$\approx 173.21 \text{ m}$$

d)

$$\dot{x}(5) = 20\sqrt{3}$$

$$\dot{y}(5) = -50 + 20 = -30$$

$$V = \sqrt{\left(20\sqrt{3}\right)^2 + (-30)^2} = 10\sqrt{21}$$

$$\approx 45.83 \text{ m/s}$$

$$\theta = \tan^{-1}\left(\frac{-30}{20\sqrt{3}}\right) = -40.89°$$

45.83 m/s at 40.89° below horizontal

e)

Let $\dot{y} = 0$

$$0 = -10t + 20$$

$$t = 2$$

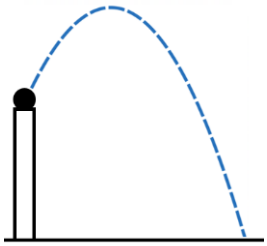
$y_{\max} = -5(2)^2 + 20(2) = 20$ above cliff

∴ max height is 20 + 25 = 45 m above ground

Example Solve projectile motion problems projected from a platform

Some questions **require** you to treat the platform as above the origin.

A stone is thrown from a point 60 m **above the origin** with speed 25 m/s at angle θ, where $\tan\theta = \frac{4}{3}$.



a Show that the displacement vector is $\underset{\sim}{r} = \begin{pmatrix} 15t \\ -5t^2 + 20t + 60 \end{pmatrix}$

b Find the time the ball takes to hit the ground at the base of the cliff.

c Find the horizontal distance from the base of the cliff to where the ball lands.

a)

$\dot{x}(0) = 25\cos\left(\tan^{-1}\left(\frac{4}{3}\right)\right) = 15$

$\dot{y}(0) = 25\sin\left(\tan^{-1}\left(\frac{4}{3}\right)\right) = 20$

$\underset{\sim}{a} = \begin{pmatrix} 0 \\ -10 \end{pmatrix}$

$\underset{\sim}{v} = \begin{pmatrix} C_1 \\ -10t + C_2 \end{pmatrix}$

Using $\underset{\sim}{v}(0) = \begin{pmatrix} 15 \\ 20 \end{pmatrix} = \begin{pmatrix} C_1 \\ -10(0) + C_2 \end{pmatrix}$

$\begin{pmatrix} 15 \\ 20 \end{pmatrix} = \begin{pmatrix} C_1 \\ C_2 \end{pmatrix}$

$\therefore \underset{\sim}{v} = \begin{pmatrix} 15 \\ -10t + 20 \end{pmatrix}$

$\underset{\sim}{r} = \begin{pmatrix} 15t + C_3 \\ -5t^2 + 20t + C_4 \end{pmatrix}$

Using $\underset{\sim}{r}(0) = \begin{pmatrix} 0 \\ 60 \end{pmatrix} = \begin{pmatrix} 15(0) + C_3 \\ -5(0)^2 + 20(0) + C_4 \end{pmatrix}$

$\begin{pmatrix} 0 \\ 60 \end{pmatrix} = \begin{pmatrix} C_3 \\ C_4 \end{pmatrix}$

$\therefore \underset{\sim}{r} = \begin{pmatrix} 15t \\ -5t^2 + 20t + 60 \end{pmatrix}$

b)

Lands when $y = 0$

$-5t^2 + 20t + 60 = 0$

$t^2 - 4t - 12 = 0$

$(t - 6)(t + 2) = 0$

$t = 6$

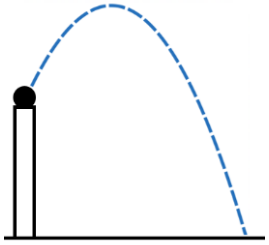
c)

$x(6) = 15 \times 6 = 90$ m

Guided Practice

A ball is thrown from the top of a 40 m building at 15 m/s at 60° above the horizontal.

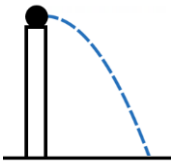
Find the time to hit the ground, range, and maximum height to 1 d.p.



Time of flight: 4.4 s
Range: 33.1 m
Max height: 48.4 m

A ball is rolled off horizontally from the top of a 25 m cliff at 20 m/s.

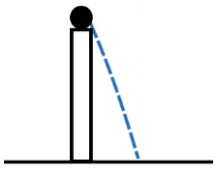
Find the time to hit the ground, range, and impact velocity to 1 d.p.



Time of flight: 2.2 s
Range: 44.7 m
Impact velocity: 30 m/s at 48.2° below horizontal

A ball is thrown from the top of a 20 m cliff at 10 m/s at 30° below the horizontal.

Find its velocity when the ball is 5 m above the ground to 1 d.p.



20 m/s at 64.3° below horizontal

Let $g = 10$ m/s² unless otherwise stated.

1 A horizontal drainpipe 6 m above sea level empties storm water horizontally into the sea. The water reaches the sea 2 m from the pipe. Find the initial speed to 1 d.p.

$\theta = 0°; y = -5t^2, x = Vt$

at $y = -6$:

$t = \sqrt{\frac{6}{5}}$

$V = \frac{2}{\sqrt{\frac{6}{5}}} \approx 1.8 \text{ m s}^{-1}$

2 A waterfall flows at 2 m s^{-1} over a vertical 5 m cliff. How far out from the cliff does it fall?

$x = 2t, y = -5t^2$

Let $y = -5$

$-5t^2 = -5$

$t_{\text{flight}} = 1$

Range = $x(1) = 2$ m

3 Jamelle throws her keys horizontally from a window 6.2 m above ground. Her friend Rochelle stands 10.4 m from the base of the building. Find the speed at which Jamelle must throw, to 1 d.p.

$x = Vt$

$y = -5t^2$

Let $y = -6.2$

$-5t^2 = -6.2$

$\therefore t = \sqrt{\frac{6.2}{5}}$ s

sub $t = \sqrt{\frac{6.2}{5}}$ and $x = 10.4$ into $x(t)$

$10.4 = V\left(\sqrt{\sqrt{\frac{6.2}{5}}}\right)$

$V \approx 9.3 \text{ m s}^{-1}$

4 A stone is thrown from the top of an 11 m high vertical tower.
The initial speed is 12 m/s and the initial direction is 30° below the horizontal.

a Starting with the horizontal and vertical components of acceleration, show that $\dot{x} = 6\sqrt{3}$ and $\dot{y} = -10t - 6$.

$\dot{x} = 12\cos 30° = 6\sqrt{3}$

$\dot{y} = -10t - 12\sin 30° = -10t - 6$ (downward initial velocity)

b If the origin is at the point of projection, find x and y in terms of t.

$x = 6\sqrt{3}t, y = -5t^2 - 6t$

c How long does it take for the stone to hit the ground?

$y = -11$

$5t^2 + 6t - 11 = 0$

$(5t + 11)(t - 1) = 0$

$t = 1\,s$

d How far from the base of the tower does it land (to 1 d.p.)?

$x(1) = 6\sqrt{3} \approx 10.4$ m

e Show the stone hits the ground at an angle of $\tan^{-1}\left(\frac{8\sqrt{3}}{9}\right)$ below the horizontal.

$\dot{x}(1) = 6\sqrt{3}$

$\dot{y}(1) = -10 - 6 = -16$

$\tan\theta = \frac{16}{6\sqrt{3}} = \frac{8}{3\sqrt{3}} = \frac{8\sqrt{3}}{9}$

$\therefore \theta = \tan^{-1}\left(\frac{8\sqrt{3}}{9}\right)$ below horizontal

5 Michele throws a frisbee from a window 15.3 m up.
The initial speed is 8.8 m s^{-1} at an angle of 73°.
Find the time taken to reach the ground.

Origin at window; $\dot{x} = 8.8\cos 73°, \dot{y} = -10t + 8.8\sin 73°$;

$y = -5t^2 + 8.8\sin 73°\, t$

Sub $y = -15.3$:

$5t^2 - 8.8\sin 73°\, t - 15.3 = 0$

$t = \frac{8.8\sin 73° + \sqrt{(8.8\sin 73°)^2 + 4(5)(15.3)}}{10} \approx 2.8\,s$

6 Origin must be at ground level

A ball is thrown from a point 50 m above the ground at 25 m/s at $\tan^{-1}\left(\frac{3}{4}\right)$ above the horizontal.

Taking the origin at ground level below the launch point, find

a x and y as functions of t,

b the time to land,

c the range from the base.

$\cos\theta = \frac{4}{5},\ \sin\theta = \frac{3}{5};\ \dot{x} = 20,\ \dot{y}(0) = 15$

a) $x = 20t;$

$y = -5t^2 + 15t + C,$

$y(0) = 50$

$y = -5t^2 + 15t + 50$

b) $y = 0:\ 5t^2 - 15t - 50 = 0$

$t^2 - 3t - 10 = 0$

$(t - 5)(t + 2) = 0$

$t = 5$ s

c) $x(5) = 100$ m

7 A ball is rolled horizontally off the top of a 45 m cliff at 20 m/s. Taking the origin at the base, find

a x and y as functions of t,

b the time to land,

c the range,

d the impact speed and angle to 1 d.p.

a) $\theta = 0:\ \dot{x} = 20,\ \dot{y}(0) = 0;$

$x = 20t,\ y = -5t^2 + 45$

b) $y = 0:\ 5t^2 = 45$

$t = 3$ s

c) $x(3) = 60$ m

d) $\dot{x} = 20,\ \dot{y}(3) = -30;$

$V = \sqrt{400 + 900} = 10\sqrt{13} \approx 36.1$ m/s,

$\theta = \tan^{-1}\left(\frac{30}{20}\right) \approx 56.3°$ below

8 A ball is thrown from a window 16 m above the ground.
The angle of projection is 60°, initial speed is 5 m s^{-1}. Find:

a the time taken to land (to 1 d.p.),

b how far from the base of the building it lands (to 1 d.p.).

a) Origin at window; $\dot{x} = 5\cos 60° = 2.5$,

$\dot{y} = -10t + 5\sin 60° = -10t + 2.5\sqrt{3}$

$x = 2.5t, y = -5t^2 + 2.5\sqrt{3}t$; ground at $y = -16$:

$5t^2 - 2.5\sqrt{3}t - 16 = 0$

$10t^2 - 5\sqrt{3}t - 32 = 0$

$t = \dfrac{5\sqrt{3} + \sqrt{75 + 1280}}{20} \approx 2.3$ s

b) $x(2.3) \approx 5.7$ m

9 A ball is tossed with initial speed 6 m/s at an angle of 45° to the horizontal. It hits the ground at a point 2 m below the point of projection. Find, to 2 d.p.:

a the time of flight,

b the horizontal distance travelled.

$\dot{x} = 6\cos 45° = 3\sqrt{2}, \dot{y} = -10t + 3\sqrt{2}$

$y = -5t^2 + 3\sqrt{2}t$

a) $y = -2$: $5t^2 - 3\sqrt{2}t - 2 = 0$

$t = \dfrac{3\sqrt{2} \pm \sqrt{18 + 40}}{10}$

$= \dfrac{3\sqrt{2} + \sqrt{58}}{10}$ $(t > 0)$

≈ 1.19 s

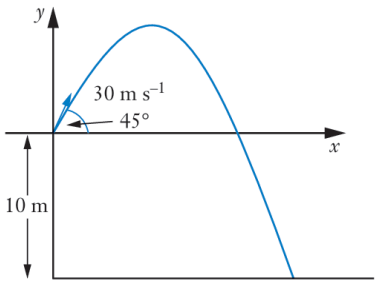
b) $x = 3\sqrt{2}\left(\dfrac{3\sqrt{2} + \sqrt{58}}{10}\right)$

$= \dfrac{9 + 3\sqrt{29}}{5} \approx 5.03$ m

10 A ball is thrown at 45° at 30 m s^{-1} and lands in a 10 m deep ditch. Using g = 9.8 m s^{-2}, find:

a how long it takes to land,

b how far out it lands.



a) $\theta = 45°, V = 30, g = 9.8$; $\dot{x} = 15\sqrt{2}, \dot{y} = -9.8t + 15\sqrt{2}$

Ground at $y = -10$: $4.9t^2 - 15\sqrt{2}\,t - 10 = 0$

$t = \dfrac{15\sqrt{2} + \sqrt{450 + 196}}{9.8} \approx 4.8$ s

b) $x = 15\sqrt{2}(4.758) \approx 100.9$ m

11 Stone is thrown from the top of a 25 m building at 30 m/s at $30°$ above the horizontal.

a Taking the origin at the base, find the time to hit the ground to 2 d.p.

b Repeat with the origin at the launch point and confirm the same time.

$\dot{x} = 30\cos 30° = 15\sqrt{3},\ \dot{y}(0) = 30\sin 30° = 15$

a) $y = -5t^2 + 15t + 25 = 0$

$t^2 - 3t - 5 = 0$

$t = 3 + \frac{\sqrt{29}}{2} \approx 4.19$ s

b) $y = -5t^2 + 15t;$

land $y = -25$:

$-5t^2 + 15t = -25$

$t^2 - 3t - 5 = 0$

$t \approx 4.19$ s

12 From the origin at a window, a ball has $x = 24t$ and $y = -5t^2 + 10t$.
It lands on the ground 4 m below the window.

a Find the time of flight (exact).

b Find how far from the base it lands to 1 d.p.

c Find the impact speed (exact).

a) $y = -4$: $-5t^2 + 10t = -4$

$5t^2 - 10t - 4 = 0$

$t = 1 + \frac{\sqrt{180}}{10} = 1 + \frac{3\sqrt{5}}{5}$ s

b) $t \approx 2.34;\ x = 24(2.34) \approx 56.2$ m

c) $\dot{x} = 24,\ \dot{y} = 10 - 10t;$

$\dot{y}^2 = 100(1-t)^2 = 180$ (from $5t^2 - 10t = 4$)

$V = \sqrt{576 + 180} = 6\sqrt{21} \approx 27.5$ m/s

13 A particle is projected from the origin at 60° with speed v. It reaches the ground after 5.1 s. Find v to 2 d.p. Use g = 9.8 m s⁻².

$$x = \frac{v}{2}t$$

$$y = -4.9t^2 + \frac{v\sqrt{3}}{2}t$$

At $t = 5.1, y = 0$

$$0 = -4.9(5.1)^2 + \frac{v\sqrt{3}}{2}(5.1)$$

After rearranging: $v = 28.86 \text{ m s}^{-1}$

14 A particle is projected from the origin with initial speed V at an angle of α to the horizontal. Two seconds later it passes through the point with position vector $8\boldsymbol{i} - 12\boldsymbol{j}$.

a Show that $V\cos\alpha = 4$ and $V\sin\alpha = 4$, and hence write down the initial velocity vector.

$$x(2) = 2V\cos\alpha = 8$$

$$V\cos\alpha = 4$$

$$y(2) = -20 + 2V\sin\alpha = -12$$

$$V\sin\alpha = 4$$

$$V = \sqrt{16 + 16} = 4\sqrt{2}, \alpha = 45°$$

$$\text{initial velocity} = 4\boldsymbol{i} + 4\boldsymbol{j}$$

b Find the position vector after 0.5 seconds.

$$\underset{\sim}{r}(0.5) = 0.5(4)\boldsymbol{i} + \big(0.5(4) - 5(0.25)\big)\boldsymbol{j}$$

$$= 2\boldsymbol{i} + 0.75\boldsymbol{j}$$

15 A particle is projected from a point O with initial speed V m/s at an angle of θ to the horizontal. After 2 seconds its horizontal and vertical displacements from O are both 30 m.

a Show that $V\cos\theta = 15$ and $V\sin\theta = 25$.

$$x(2) = 2V\cos\theta = 30$$

$$V\cos\theta = 15$$

$$y(2) = -20 + 2V\sin\theta = 30$$

$$V\sin\theta = 25$$

b Hence show that $V = 5\sqrt{34}\text{ m/s}$ and $\theta = \tan^{-1}\left(\frac{5}{3}\right)$.

$$V = \sqrt{225 + 625} = 5\sqrt{34}$$

$$\tan\theta = \frac{25}{15} = \frac{5}{3}$$

$$\theta = \tan^{-1}\left(\frac{5}{3}\right)$$

16 Points A and B lie on a vertical line above the origin O on the ground, with $OA = kh$ and $OB = h$ where $k > 1$.
A particle is projected horizontally from A with speed U. Some time later, a second particle is projected horizontally from B with speed V. Both land at the same point on the ground.
Show that $\frac{V}{U} = \sqrt{k}$.

A: $x_A = Ut,\ y_A = kh - \frac{1}{2}gt^2$

land $y_A = 0$

$t = \sqrt{\frac{2kh}{g}}$

$x_A = U\sqrt{\frac{2\text{kh}}{\text{g}}}$

B (let time be T): $y_B = h - \frac{1}{2}gT^2 = 0$

$T = \sqrt{\frac{2h}{g}}$

$x_B = V\sqrt{\frac{2\text{h}}{\text{g}}}$

Same landing point: $U\sqrt{\frac{2kh}{g}} = V\sqrt{\frac{2\text{h}}{\text{g}}}$

$\frac{V}{U} = \sqrt{\frac{\left(\frac{2kg}{h}\right)}{\left(\frac{2h}{g}\right)}} = \sqrt{k}$

17 A projectile is launched from a point at height h above the ground (origin at the launch point) with speed V at angle θ. Show it lands after $t = \frac{V\sin\theta + \sqrt{V^2\sin^2\theta + 2gh}}{g}$.

$a)\ y = -\frac{1}{2}gt^2 + V\sin\theta\, t;$

land $y = -h$: $\frac{1}{2}gt^2 - V\sin\theta\, t - h = 0$

$gt^2 - 2V\sin\theta\, t - 2gh = 0$

$t = \frac{2V\sin\theta + \sqrt{4V^2\sin^2\theta + 8gh}}{2g}$

$= \frac{V\sin\theta + \sqrt{V^2\sin^2\theta + 2gh}}{g}$

18 A ball was projected from ground level and, when at its highest point, just cleared a 3 m wall. Given the initial velocity was 20 m/s at angle α to the horizontal, prove that $\alpha = \sin^{-1}\left(\frac{\sqrt{15}}{10}\right)$.

At the highest point, vertical velocity = 0.

Vertical component: $\dot{y} = 20\sin\alpha - 10t = 0$

$$t = \frac{20\sin\alpha}{10}$$

Height at highest point = 3:

$$y = -5t^2 + 20\sin\alpha\, t$$

$$-5\left(\frac{20\sin\alpha}{10}\right)^2 + 20\sin\alpha\left(\frac{20\sin\alpha}{10}\right) = 3$$

$$-20\sin^2\alpha + 40\sin^2\alpha = 3$$

$$\sin^2\alpha = \frac{3}{20}$$

$$\sin\alpha = \frac{\sqrt{3}}{\sqrt{20}} = \frac{\sqrt{3}}{2\sqrt{5}} = \frac{\sqrt{15}}{10}$$

$$\therefore \alpha = \sin^{-1}\left(\frac{\sqrt{15}}{10}\right)$$

19 Two particles P_1 and P_2 are projected from the origin with initial velocity vectors $20\boldsymbol{i} + 30\boldsymbol{j}$ and $60\boldsymbol{i} + 50\boldsymbol{j}$ respectively. P_2 is projected 2 seconds after P_1.
Determine whether they collide and, if so, when.

$$P_1: x_1 = 20t, y_1 = -5t^2 + 30t$$

$$P_2(t \geq 2): x_2 = 60(t-2), y_2 = -5(t-2)^2 + 50(t-2)$$

Let $x_1 = x_2$:

$$20t = 60t - 120$$

$$t = 3$$

Check y values:

$$y_1(3) = -45 + 90 = 45$$

$$y_2(3) = -5 + 50 = 45$$

They collide at $t = 3$ s (1 s after P_2 is projected).

20 A particle is projected from level ground with initial speed V m/s at an angle of θ to the horizontal.

a Prove that:

i the greatest height is $\frac{V^2\sin^2\theta}{2g}$ metres,

ii the horizontal range is $\frac{V^2\sin 2\theta}{g}$ metres.

b If the horizontal range is five times the greatest height, prove that $\theta = \tan^{-1}\left(\frac{4}{5}\right)$

a i)

At greatest height, $\dot{y} = 0$:

$$V\sin\theta - gt = 0$$

$$t = \frac{V\sin\theta}{g}$$

$$H = V\sin\theta \cdot \frac{V\sin\theta}{g} - \frac{1}{2}g\left(\frac{V\sin\theta}{g}\right)^2$$

$$= \frac{V^2\sin^2\theta}{g} - \frac{V^2\sin^2\theta}{2g}$$

$$= \frac{V^2\sin^2\theta}{2g}$$

a ii)

Lands when $y = 0$:

$$V\sin\theta \cdot t - \frac{1}{2}gt^2 = 0$$

$$t\left(V\sin\theta - \frac{1}{2}gt\right) = 0$$

$$t = \frac{2V\sin\theta}{g}$$

$$R = V\cos\theta \cdot \frac{2V\sin\theta}{g} = \frac{V^2(2\sin\theta\cos\theta)}{g}$$

$$= \frac{V^2\sin 2\theta}{g}$$

b)

$R = 5H$:

$$\frac{V^2\sin 2\theta}{g} = 5 \cdot \frac{V^2\sin^2\theta}{2g}$$

$$2\sin\theta\cos\theta = \frac{5}{2}\sin^2\theta$$

Divide both sides by sin θ (sin θ ≠ 0):

$$2\cos\theta = \frac{5}{2}\sin\theta$$

$$\tan\theta = \frac{4}{5}$$

$$\therefore \theta = \tan^{-1}\left(\frac{4}{5}\right)$$

Further Problems with Projectile Motion

Review

☐ Rearrange equation for t

a $60 = 30\cos\theta\, t$

b $x = V\cos\theta\, t$

☐ Simplify expression using trigonometric identities

a Simplify.

$$30\sin\theta \cdot \frac{2}{\cos\theta}$$

b Write in terms of $\sec\theta$

$$5\left(\frac{2}{\cos\theta}\right)^2$$

☐ Use the identity $\sec^2\theta = 1 + \tan^2\theta$

a Write in terms of $\tan\theta$

$$6\tan\theta - \sec^2\theta = 4$$

b Write in terms of $\tan\theta$

$$40\tan\theta - 20\sec^2\theta = 20$$

☐ Solve a quadratic in $\tan\theta$

a $\tan^2\theta - 3\tan\theta + 2 = 0$

b $4\tan^2\theta - 20\tan\theta + 9 = 0$

ⓘ Solving for t and using it to show a result

When a projectile passes through a known point and the angle or speed is unknown:

0. Derive equations of motion, if not already given.

1. Solve the horizontal equation for t (since it is linear in t)

2. Substitute into the vertical equation.

3. Use $\sec^2\theta = 1 + \tan^2\theta$ to obtain a quadratic in $\tan\theta$.

Example Solve further problems involving projectile motion

A particle is projected from the origin with speed 30 m/s at angle θ.

It is given that $\underset{\sim}{r} = \begin{pmatrix} 30\cos\theta\, t \\ -5t^2 + 30\sin\theta\, t \end{pmatrix}$

If it passes through the point $(60,20)$, find two possible angles of projection.

1. Solve horizontal equation for t

$x = 30\cos\theta\, t$

Let $x = 60$:

$60 = 30\cos\theta\, t$

$t = \dfrac{2}{\cos\theta}$

2. Substitute into vertical equation

$y = -5t^2 + 30\sin\theta\, t$

Let $y = 20, t = \dfrac{2}{\cos\theta}$

$20 = -5\left(\dfrac{2}{\cos\theta}\right)^2 + 30\sin\theta\left(\dfrac{2}{\cos\theta}\right)$

$20 = -20\sec^2\theta + 30\tan\theta$

3. Quadratic in $\tan\theta$.

$20 = -20(1 + \tan^2\theta) + 60\tan\theta$

$20 = -20 - 20\tan^2\theta + 60\tan\theta$

$20\tan^2\theta - 60\tan\theta + 40 = 0$

$\tan^2\theta - 3\tan\theta + 2 = 0$

$(\tan\theta - 1)(\tan\theta - 2) = 0$

$\tan\theta = 1$ or $\tan\theta = 2$

$\theta = 45°$ or $\theta = \tan^{-1}(2) \approx 63.4°$

Guided Practice

A cannon is fixed at an angle of 60° but can fire cannonballs at variable initial velocity V.

Find a possible initial velocity at which the cannon can be set so that it can hit a castle at a horizontal distance of 2000 m away at a top of a 1000 m high mountain to 2 d.p.

https://www.desmos.com/calculator/gg3adnbr5n

180.18 m/s

A cannon at ground level fires shells with a fixed initial speed 200 m/s and variable angle α.

Find two possible angles at which the cannon can be set so that it can hit a castle at a horizontal distance of 2000 m away at a top of a 1000 m high mountain to 2 d.p.

https://www.desmos.com/calculator/ezmt9ar5rk

45° and 71° 34'

1 A particle is projected from the origin at 30 m/s and passes through $(30,20)$.

It is given that $\underset{\sim}{r} = \begin{pmatrix} 30\cos\theta\, t \\ -5t^2 + 30\sin\theta\, t \end{pmatrix}$

Show that $\tan^2\theta - 6\tan\theta + 5 = 0$ and find both possible angles.

$x = 30\cos\theta\, t = 30$

$t = \frac{1}{\cos\theta}$

$y = 20\text{: } 30\sin\theta \cdot \frac{1}{\cos\theta} - 5\left(\frac{1}{\cos\theta}\right)^2 = 20$

$30\tan\theta - 5\sec^2\theta = 20$

$30\tan\theta - 5(1 + \tan^2\theta) = 20$

$5\tan^2\theta - 30\tan\theta + 25 = 0$

$\tan^2\theta - 6\tan\theta + 5 = 0$

$(\tan\theta - 1)(\tan\theta - 5) = 0$

$\theta = 45°$ or $\tan^{-1}5 \approx 78.7°$

2 A particle is projected from the origin at 50 m/s and passes through $(100,25)$.

a Find the displacement vector by integration.

b Show that $4\tan^2\theta - 20\tan\theta + 9 = 0$ and find both angles.

3 A projectile was fired from level ground and just cleared a wall 10 m high and 20 m from the point of projection. If the angle of projection was 36°, find the initial speed to the nearest integer.
Note: the height of the wall is not necessarily the maximum height.

$$\text{Let } \alpha = 36°, \text{horizontal distance } x = 20, \text{vertical height } y = 10.$$

$$x = V\cos\alpha \cdot t$$

$$y = V\sin\alpha \cdot t - \frac{1}{2}(10)t^2$$

$$\text{From } x = 20: t = \frac{20}{V\cos 36°}$$

$$\text{Sub into } y = 10:$$

$$10 = V\sin 36° \cdot \left(\frac{20}{V\cos 36°}\right) - 5\left(\frac{20}{V\cos 36°}\right)^2$$

$$10 = 20\tan 36° - \frac{5(400)}{V^2\cos^2 36°}$$

$$\frac{5(400)}{V^2\cos^2 36°} = 20\tan 36° - 10$$

$$V^2 = \frac{2000}{\cos^2 36°(20\tan 36° - 10)}$$

$$V^2 \approx 674.5$$

$$V \approx 26 \text{ m/s}$$

4 A ball is thrown from the top of a 15 m tower.
Taking the origin at the base, it is projected at 20 m/s and lands 40 m from the base.
Find the two possible angles to 1 d.p.

$$x = 20\cos\theta\, t = 40$$

$$t = \frac{2}{\cos\theta}$$

$$y = -5t^2 + 20\sin\theta\, t + 15$$

$$\text{land } y = 0: -5\left(\frac{2}{\cos\theta}\right)^2 + 20\sin\theta \cdot \frac{2}{\cos\theta} + 15 = 0$$

$$-20\sec^2\theta + 40\tan\theta + 15 = 0$$

$$-20(1 + \tan^2\theta) + 40\tan\theta + 15 = 0$$

$$20\tan^2\theta - 40\tan\theta + 5 = 0$$

$$4\tan^2\theta - 8\tan\theta + 1 = 0$$

$$\tan\theta = 8 \pm \frac{\sqrt{48}}{8}$$

$$= 1 \pm \frac{\sqrt{3}}{2}$$

$$\theta \approx 61.8° \text{ or } 7.6°$$

5 A particle is projected with initial speed 50 m/s. On its flight it passes through a point P at horizontal distance 100 m and vertical distance 25 m from the point of projection.

a Show that the angle of projection α is $\tan^{-1}\left(\frac{1}{2}\right)$ or $\tan^{-1}\left(\frac{9}{2}\right)$.

b For each angle, find:

i the time to reach P,

ii the velocity at P (to 1 d.p. where necessary).

a) $x = 50\cos\alpha\, t = 100$

$$t = \frac{2}{\cos\alpha}$$

Sub into $y = 50\sin\alpha\, t - 5t^2 = 25$:

$$50\sin\alpha \cdot \left(\frac{2}{\cos\alpha}\right) - 5\left(\frac{4}{\cos^2\alpha}\right) = 25$$

$$100\tan\alpha - 20\sec^2\alpha = 25$$

$$100\tan\alpha - 20(1 + \tan^2\alpha) = 25$$

$$20\tan^2\alpha - 100\tan\alpha + 45 = 0$$

$$4\tan^2\alpha - 20\tan\alpha + 9 = 0$$

$$(2\tan\alpha - 1)(2\tan\alpha - 9) = 0$$

$$\tan\alpha = \frac{1}{2} \text{ or } \tan\alpha = \frac{9}{2}$$

$$\therefore \alpha = \tan^{-1}\left(\frac{1}{2}\right) \text{ or } \alpha = \tan^{-1}\left(\frac{9}{2}\right)$$

b)

$\alpha = \tan^{-1}\left(\frac{1}{2}\right)$: $\sin\alpha = \frac{1}{\sqrt{5}}, \cos\alpha = \frac{2}{\sqrt{5}}$

i) $t = \frac{2}{\cos\alpha} = \frac{2}{\frac{2}{\sqrt{5}}} = \sqrt{5} \approx 2.2\text{ s}$

ii) $\dot{x} = 50\cos\alpha = \frac{100}{\sqrt{5}} = 20\sqrt{5}$

$\dot{y} = 50\sin\alpha - 10t = \frac{50}{\sqrt{5}} - 10\sqrt{5} = 10\sqrt{5} - 10\sqrt{5} = 0$

Speed = $20\sqrt{5} \approx 44.7$ m/s , directed horizontally.

(P is the maximum height)

$\alpha = \tan^{-1}\left(\frac{9}{2}\right)$: $\sin\alpha = \frac{9}{\sqrt{85}}, \cos\alpha = \frac{2}{\sqrt{85}}$

i) $t = \frac{2}{\cos\alpha} = \frac{2}{\frac{2}{\sqrt{85}}} = \sqrt{85} \approx 9.2\,s$

ii) $\dot{x} = 50\cos\alpha = \frac{100}{\sqrt{85}}$

$\dot{y} = 50\sin\alpha - 10t = \frac{450}{\sqrt{85}} - 10\sqrt{85} = \frac{450 - 850}{\sqrt{85}} = -\frac{400}{\sqrt{85}}$

Speed $= \sqrt{\dot{x}^2 + \dot{y}^2} = \sqrt{\frac{100^2 + 400^2}{85}} = 20\sqrt{5} \approx 44.7$ m/s

Direction: $\theta = \tan^{-1}\left(\frac{400}{100}\right) = \tan^{-1}(4)$ below horizontal $\approx 76.0°$ below horizontal.

6 A particle is projected from the origin with initial speed $20\sqrt{2}$ m/s.
On its flight it passes through the point (20, 15) at time t. Let θ be the angle of projection.

a Show that $\sqrt{2}t\cos\theta = 1$ and $4\sqrt{2}\,t\sin\theta - t^2 = 3$.

b Hence show that $\tan^2\theta - 8\tan\theta + 7 = 0$.

c Hence find, to the nearest minute where necessary, the two possible values of θ.

$a)\ x = 20\sqrt{2}t\cos\theta = 20$

$\sqrt{2}t\cos\theta = 1$

$y = -5t^2 + 20\sqrt{2}t\sin\theta = 15$

$4\sqrt{2}t\sin\theta - t^2 = 3$

b) From $\sqrt{2}t\cos\theta = 1$

$t = \frac{1}{\sqrt{2}\cos\theta}$

Sub into $4\sqrt{2}t\sin\theta - t^2 = 3$:

$4\tan\theta - \frac{1}{2}\sec^2\theta = 3$

$8\tan\theta - (1 + \tan^2\theta) = 6$

$\tan^2\theta - 8\tan\theta + 7 = 0$

c) $(\tan\theta - 1)(\tan\theta - 7) = 0$

$\tan\theta = 1$ or 7

$\theta = 45°$ or $\tan^{-1}(7) \approx 81°52'$

7 Cooper kicks a football at 12 m s^{-1} and just clears a goalpost 4 m high and 9 m away.
Find the angle of projection to the nearest degree.

$x = 12\cos\theta\, t$

$y = 12\sin\theta\, t - 5t^2$

Sub in (9,4)

$9 = 12\cos\theta\, t\ (1)$

$4 = 12\sin\theta\, t - 5t^2\ (2)$

From (1): $t = \frac{3}{4\cos\theta}$

Sub into (2)

$12\sin\theta \cdot \left(\frac{3}{4\cos\theta}\right) - 5\left(\frac{3}{4\cos\theta}\right)^2 = 4$

$\frac{9\tan\theta}{1} - \frac{45}{16\cos^2\theta} = 4$

Using $\sec^2\theta = 1 + \tan^2\theta$:

$9\tan\theta - \left(\frac{45}{16}\right)(1 + \tan^2\theta) = 4$

$45\tan^2\theta - 144\tan\theta + 109 = 0$

Using quadratic formula, $\tan\theta = \frac{144 \pm 6\sqrt{31}}{90}$

$\theta \approx 63°$ and $51°$

Cartesian Equation of a Path

Review

☐ Differentiate a quadratic

c $y = \frac{4}{3}x - \frac{1}{72}x^2$. Find $\frac{dy}{dx}$.

d $y = \frac{1}{2}x - \frac{1}{160}x^2$. Find $\frac{dy}{dx}$.

☐ Solve $y = 0$ for a non-zero root

a $0 = \frac{4}{3}x - \frac{1}{72}x^2$

b $0 = x\left(2 - \frac{x}{50}\right)$

☐ Interpret gradient as an angle

a The gradient of a projectile's path at a point is $\frac{3}{4}$. Find the angle of the direction of motion, to 1 d.p.

b At a point the gradient of a projectile's path is -2. State the direction of motion as an angle below the horizontal.

Investigation General Cartesian equation of the path of a projectile

We have expressed the horizontal and vertical displacements in terms of time t.

This is a **parametrically** defined curve, with parameter t.

$$x = Vt\cos\theta \quad \text{and} \quad y = -\frac{1}{2}gt^2 + Vt\sin\theta$$

It would be useful to have a **Cartesian** equation that relates x and y directly.

Eliminate the parameter t to show that

$$y = -\frac{gx^2}{2V^2}(1 + \tan^2\theta) + x\tan\theta$$

ⓘ Cartesian Equation of Projectile Motion

$$y = -\frac{gx^2}{2V^2}\sec^2\theta + x\tan\theta$$

$$y = -\frac{gx^2}{2V^2}(1 + \tan^2\theta) + x\tan\theta$$

Do not memorise these equations. They will either be given, or you will need to derive them. The Cartesian equation can be very useful as it relates x and y directly:

- The instantaneous direction of the projectile is given by $\tan\phi = \frac{dy}{dx}$.
- The range is given by setting $y = 0$.
- The maximum height can be found by setting $\frac{dy}{dx} = 0$, or by using $x = -\frac{b}{2a}$

Example Use the Cartesian equation of a projectile

A projectile launched from the origin follows the path $y = \frac{4}{3}x - \frac{1}{72}x^2$,

where x and y are in metres. Find:

a The angle of projection.
b The direction of motion at $x = 60$.
c The range.
d The maximum height.

a. angle of projection

$\frac{dy}{dx} = \frac{4}{3} - \frac{x}{36}$

at $x = 0$, $\frac{dy}{dx} = \frac{4}{3}$

$\tan\theta = \frac{4}{3}$

$\theta = \tan^{-1}\left(\frac{4}{3}\right) \approx 53.1°$

b. direction of motion at $x = 60$

at $x = 60$, $\frac{dy}{dx} = \frac{4}{3} - \frac{60}{36} = -\frac{1}{3}$

$\tan^{-1}\left(\frac{1}{3}\right) \approx 18.4°$ below the horizontal

c. range

Let $y = 0$

$\frac{4}{3}x - \frac{1}{72}x^2 = 0$

$x\left(\frac{4}{3} - \frac{1}{72}x\right) = 0$

$x = 0,\ x = 96$

∴ range is 96 m

d. maximum height

Let $\frac{dy}{dx} = 0$

$\frac{4}{3} - \frac{x}{36} = 0$

$\frac{x}{36} = \frac{4}{3}$

$x = 48$

$y(48) = \frac{4}{3}(48) - \frac{1}{72}(48)^2 = 32$ m

Guided Practice

A projectile from the origin follows $y = \frac{1}{2}x - \frac{1}{160}x^2$.

Find the angle of projection, the range, and the maximum height.

Angle of projectile: $\tan^{-1}\frac{1}{2}$
Range: 80
Max height: 10

A projectile has parametric equations $x = 48t,\ y = -5t^2 + 20t$.

Find the Cartesian equation, then use it to find the range.

$y = -\frac{5x^2}{2304} + \frac{5x}{12}$
Range: 192 m

1 A projectile from the origin follows $y = \frac{1}{2}x - \frac{1}{160}x^2$. Find

a the angle of projection,

b the range,

c the greatest height.

2 A projectile from the origin follows $y = \frac{5}{3}x - \frac{5}{216}x^2$. Find

a the angle of projection to 1 d.p.,

b the range,

c the greatest height.

3 A projectile from the origin has parametric equations $x = 20t$, $y = -5t^2 + 30t$.

a Show the Cartesian equation is $y = \frac{3}{2}x - \frac{1}{80}x^2$.

b Find the range and greatest height.

c Find the direction of motion when $x = 100$.

4 A particle is projected from a point O.
Its parabolic motion is governed by $x = 48t$ and $y = -5t^2 + 20t$, where x and y are the horizontal and vertical components of displacement from O after t seconds.

a Show that the Cartesian equation is $y = -\frac{5}{2304}x^2 + \frac{5}{12}x$.

$$t = \frac{x}{48};\ y = -5\left(\frac{x}{48}\right)^2 + 20\left(\frac{x}{48}\right)$$
$$= -\frac{5}{2304}x^2 + \frac{5}{12}x$$

b Use the Cartesian equation to find:

i the horizontal range,

ii the greatest height,

iii the angle of projection (to 1 d.p.),

iv the direction of motion when $x = 120$ (to the nearest degree).

i) $y = 0\ (x > 0)$: $\left(\frac{5}{12}\right)x = \left(\frac{5}{2304}\right)x^2$

$$x = 192 \text{ m}$$

ii) $\frac{dy}{dx} = 0$: $-\frac{10x}{2304} + \frac{5}{12} = 0$

$$x = 96$$

$$y(96) = -\frac{5(9216)}{2304} + \frac{5(96)}{12} = 20 \text{ m}$$

iii) $\frac{dy}{dx}$ at $x = 0$: $\frac{5}{12}$

$$\theta = \tan^{-1}\left(\frac{5}{12}\right) \approx 22.6°$$

iv) At $x = 120$: $\frac{dy}{dx} = -\frac{600}{2304} + \frac{5}{12} = -\frac{5}{48}$

$$\theta = \tan^{-1}\left(\frac{5}{48}\right) \approx 6° \text{ below horizontal}$$

5 A particle is projected from the origin and its parabolic path has equation $y = -\frac{5}{324}x^2 + \frac{4}{3}x$.

a Find the height when the particle has travelled a horizontal distance of 12 m.

$y(12) = -\left(\frac{5}{324}\right)(144) + \left(\frac{4}{3}\right)(12) = \frac{124}{9} \approx 13.8 \text{ m}$

b Find how far the particle has travelled horizontally when its height is 19 m.

$-\frac{5}{324}x^2 + \frac{4}{3}x = 19$

multiply by 324: $5x^2 - 432x + 6156 = 0$

Using quadratic formula:

$x = \frac{432 \pm 252}{10} \rightarrow x = 18 \text{ m or } 68.4 \text{ m}$

c Find $\frac{dy}{dx}$ and hence show that:

i the angle of projection is $\tan^{-1}\left(\frac{4}{3}\right)$,

ii when $x = 18$ m, the direction of motion is $\tan^{-1}\left(\frac{7}{9}\right)$ above the horizontal,

iii when $x = 54$ m, the direction of motion is $\tan^{-1}\left(\frac{1}{3}\right)$ below the horizontal.

$\frac{dy}{dx} = -\frac{5}{162}x + \frac{4}{3}$

i) At $x = 0$: $\frac{dy}{dx} = \frac{4}{3}$

$\theta = \tan^{-1}\left(\frac{4}{3}\right)$

ii) At $x = 18$:

$\frac{dy}{dx} = -\frac{90}{162} + \frac{4}{3} = \frac{7}{9}$

$\tan^{-1}\left(\frac{7}{9}\right)$ above horizontal

iii) At $x = 54$:

$\frac{dy}{dx} = -\frac{270}{162} + \frac{4}{3} = -\frac{1}{3}$

$\tan^{-1}\left(\frac{1}{3}\right)$ below horizontal

6 A projectile from the origin follows $y = x\tan\theta - \frac{5x^2}{V^2}\sec^2\theta$. It is launched at $V = 20$ m/s and passes through $(20,10)$.

a Show that $\tan^2\theta - 4\tan\theta + 3 = 0$.

b Find the two possible angles.

a) $\frac{5x^2}{V^2} = \frac{5(400)}{400} = 5;$

$10 = 20\tan\theta - 5\sec^2\theta$

$10 = 20\tan\theta - 5(1 + \tan^2\theta)$

$5\tan^2\theta - 20\tan\theta + 15 = 0$

$\tan^2\theta - 4\tan\theta + 3 = 0$

b) $(\tan\theta - 1)(\tan\theta - 3) = 0$

$\theta = 45°$ or $\tan^{-1}3 \approx 71.6°$

7 An object is rolled horizontally from the top of a 20 m tower with initial velocity vector $5\boldsymbol{i}$.

a If the point of projection is the origin, find the Cartesian equation of the path.

b Find how far the object lands from the base of the tower.

c Find, to the nearest degree, the direction of motion when it hits the ground.

a) $\dot{x} = 5, \dot{y} = -10t$

$x = 5t, y = -5t^2$

$t = \frac{x}{5}$

$y = -\frac{x^2}{5}$

b) $y = -20: x^2 = 100$

$x = 10$ m from base

c) At $y = -20: t = 2;\ \dot{x} = 5, \dot{y} = -20$

$\theta = \tan^{-1}\left(\frac{20}{5}\right) = \tan^{-1}(4) \approx 76°$ below horizontal

8 A particle is projected from the origin O with initial velocity vector $3\boldsymbol{i} + \boldsymbol{j}$.

a What is the initial speed?

$\text{Speed} = \sqrt{9 + 1} = \sqrt{10}\ \text{m/s}$

b Show that the parabolic path has parametric equations $x = 3t$ and $y = t - 5t^2$.

$\dot{x} = 3$

$x = 3t$

$\dot{y} = 1 - 10t$

$y = t - 5t^2$

c Hence show that the Cartesian equation is $9y = 3x - 5x^2$.

$t = \frac{x}{3} : y = \frac{x}{3} - 5\left(\frac{x}{3}\right)^2$

$9y = 3x - 5x^2$

d Find, to 1 d.p., the direction of motion when:

i $x = 0.15$

ii $t = 0.15$

$\frac{dy}{dx} = \frac{1}{3} - \frac{10x}{9}$

i) At $x = 0.15$: $\frac{dy}{dx} = \frac{1}{3} - \frac{1}{6} = \frac{1}{6}$

$\tan^{-1}\left(\frac{1}{6}\right) \approx 9.5°$ above horizontal

ii) At $t = 0.15$: $\dot{x} = 3, \dot{y} = 1 - 1.5 = -0.5$

$\frac{dy}{dx} = -\frac{0.5}{3} = -\frac{1}{6}$

9.5° below horizontal

9 A ball is tossed from the origin O that is 6 m above ground level.
The initial velocity is 24 m/s at an angle of 30° above the horizontal.

a Find $\dot{x}, \dot{y}, x$ and y as functions of t.

b Show that the Cartesian equation of the path is $y = \frac{1}{\sqrt{3}}x - \frac{5}{432}x^2$.

c If D metres is the horizontal distance travelled when it strikes the ground, show that

$$5D^2 - 144\sqrt{3}D - 2592 = 0$$

d Hence find D correct to 1 d.p.

a) $\dot{x} = 12\sqrt{3}, \dot{y} = -10t + 12,$

$x = 12\sqrt{3}t, y = -5t^2 + 12t$

b) $t = \frac{x}{12\sqrt{3}}:$

$y = -\frac{5x^2}{432} + \frac{x}{\sqrt{3}}$

$= \frac{1}{\sqrt{3}}x - \frac{5}{432}x^2$

c) Ground at $y = -6$: $\left(\frac{1}{\sqrt{3}}\right)D - \left(\frac{5}{432}\right)D^2 = -6;$ multiply by 432:

$5D^2 - 144\sqrt{3}D - 2592 = 0$

d) $D = \frac{144\sqrt{3} + \sqrt{62208 + 51840}}{10} \approx 58.7\ \text{m}$

10 A particle is projected from the origin with initial speed V at angle θ to the horizontal.

You may assume the equation of its path is $y = x\tan\theta - \frac{gx^2\sec^2\theta}{2V^2}$

a Show that the horizontal range is $\frac{V^2\sin 2\theta}{g}$.

b If $V = 30$ m/s and the range is to be 75 m, find the two possible values of θ to 1 d.p.

a) At $y = 0$ $(x > 0)$: $x\tan\theta = \frac{gx^2\sec^2\theta}{2V^2}$

$x = \frac{2V^2\tan\theta}{g\sec^2\theta} = \frac{2V^2\sin\theta\cos\theta}{g} = \frac{V^2\sin 2\theta}{g}$

b) $\frac{V^2\sin 2\theta}{g} = 75$:

$\frac{900\sin 2\theta}{10} = 75$

$\sin 2\theta = \frac{5}{6}$

$2\theta = \sin^{-1}\left(\frac{5}{6}\right) \approx 56.4°$ or $123.6°$

$\theta \approx 28.2°$ or $61.8°$

11 A particle is projected from the origin with initial speed V at angle θ.

You may assume the equation of its parabolic path is $y = x\tan\theta - \frac{gx^2\sec^2\theta}{2V^2}$

a Find $\frac{dy}{dx}$, and hence show that the vertex of the parabola is $\left(\frac{V^2\sin 2\theta}{2g}, \frac{V^2\sin^2\theta}{2g}\right)$.

b If $V = 20$ m/s and the greatest height is 15 m, find θ.

a) $\frac{dy}{dx} = \tan\theta - \frac{gx\sec^2\theta}{V^2}$

set to 0: $x = \frac{V^2\tan\theta}{g\sec^2\theta} = \frac{V^2\sin 2\theta}{2g}$

Sub into y equation: $y_{vertex} = \frac{V^2\sin 2\theta}{2g}\tan\theta - \left(\frac{V^2\sin 2\theta}{2g}\right)^2\frac{g\sec^2\theta}{2V^2}$

$= \frac{V^2\,2\sin\theta\cos\theta}{2g}\frac{\sin\theta}{\cos\theta} - \frac{V^4 4\sin^2\theta\cos^2\theta}{4g^2}\frac{g}{2V^2\cos^2\theta}$

$= \frac{V^2\sin^2\theta}{g} - \frac{V^2\sin^2\theta}{2g}$

$= \frac{V^2\sin^2\theta}{2g}$

b) $\frac{V^2\sin^2\theta}{2g} = 15$:

$\frac{400\sin^2\theta}{20} = 15$

$\sin^2\theta = \frac{3}{4}$

$\theta = 60°$

12 A stone is thrown from the top of a 60 m cliff at an angle of 27° below the horizontal. It lands in the ocean 35 m from the base of the cliff.

a Taking the origin at the point of projection, show that the path has Cartesian equation $y = -x\tan 27° - \frac{5x^2}{V^2\cos^2 27°}$, where V is the initial speed.

b Hence find:

i the initial speed to 1 d.p.,

ii the direction of motion when it lands, to the nearest degree.

$a)\ \dot{x} = V\cos 27°, \dot{y} = -10t - V\sin 27°;$

$t = \frac{x}{V\cos 27°}$

$y = -\frac{x\sin 27°}{\cos 27°} - \frac{5x^2}{V^2\cos^2 27°}$

$= -x\tan 27° - \frac{5x^2}{V^2\cos^2 27°}$

b)

i) At $x = 35, y = -60$:

$-60 = -35\tan 27° - \frac{6125}{V^2\cos^2 27°}$

$\frac{6125}{V^2\cos^2 27°} = 60 - 35\tan 27°$

$V^2 = \frac{6125}{(60 - 35\tan 27°) \times \cos^2 27°} \approx 183.0$

$V \approx 13.5\ \text{m/s}$

ii) $t = \frac{35}{13.5\cos 27°} \approx 2.91\ \text{s}$

$\dot{x} \approx 12.03, \dot{y} = -10(2.91) - 13.5\sin 27° \approx -35.2$

$\theta = \tan^{-1}\left(\frac{35.2}{12.03}\right) \approx 71°\ \text{below horizontal}$

13 A particle is projected from a point O with speed 34 m/s at angle θ above the horizontal.

a Show that the parabolic path has equation $y = x\tan\theta - \frac{5x^2\sec^2\theta}{1156}$

b The particle passes through a point 11 m above O and 30 m horizontally from O.
Show that the two possible angles of projection are $\theta = \tan^{-1}\left(\frac{8}{15}\right)$ and $\theta = \tan^{-1}\left(\frac{538}{75}\right)$

a) $y = x\tan\theta - \left(\frac{10x^2}{2(34^2)}\right)\sec^2\theta = x\tan\theta - \frac{5x^2\sec^2\theta}{1156}$

b) At $(30, 11)$, let $\tau = \tan\theta$:

$11 = 30\tau - \left(\frac{4500}{1156}\right)(1 + \tau^2) = 30\tau - \left(\frac{1125}{289}\right)(1 + \tau^2)$

Multiply by 289: $3179 = 8670\tau - 1125(1 + \tau^2)$

$1125\tau^2 - 8670\tau + 4304 = 0$

$\tau = \frac{8670 \pm 7470}{2250}$

$\tau = \frac{538}{75}\ \text{or}\ \frac{8}{15}$

$\therefore\ \theta = \tan^{-1}\left(\frac{8}{15}\right)\ \text{or}\ \tan^{-1}\left(\frac{538}{75}\right)$

14 A gun fires a shell with constant initial speed V and variable angle of elevation α.
The displacements after t seconds are $x = Vt\cos\alpha$ and $y = -\frac{1}{2}gt^2 + Vt\sin\alpha$.

a Show that the Cartesian equation may be written as

$$gx^2\tan^2\alpha - 2xV^2\tan\alpha + (2yV^2 + gx^2) = 0$$

b Suppose $V = 200$ m/s and the shell hits a target 3 km horizontally and 0.5 km vertically from the gun. Show that $\tan\alpha = \frac{4+\sqrt{3}}{3}$ or $\tan\alpha = \frac{4-\sqrt{3}}{3}$, and hence find the two possible values of α to the nearest minute.

a) From $x = Vt\cos\alpha$: $t = \frac{x}{V\cos\alpha}$; sub into y:

$$y = -\left(\frac{g}{2}\right)\left(\frac{x}{V\cos\alpha}\right)^2 + \frac{Vx\sin\alpha}{V\cos\alpha}$$

$$= x\tan\alpha - \frac{gx^2\sec^2\alpha}{2V^2}$$

Rearrange: $\left(\frac{gx^2}{2V^2}\right)(1 + \tan^2\alpha) - x\tan\alpha + y = 0$; multiply by $2V^2$:

$$gx^2\tan^2\alpha - 2xV^2\tan\alpha + (2yV^2 + gx^2) = 0$$

b) $V = 200, x = 3000, y = 500, g = 10$:

$$10(9\times10^6)\tan^2\alpha - 2(3000)(4\times10^4)\tan\alpha + (2(500)(4\times10^4) + 9\times10^7) = 0$$

$$9\tan^2\alpha - 24\tan\alpha + 13 = 0$$

$$\tan\alpha = \frac{24 \pm \sqrt{576-468}}{18} = \frac{24 \pm 6\sqrt{3}}{18} = \frac{4\pm\sqrt{3}}{3}$$

$$\tan\alpha = \frac{4+\sqrt{3}}{3}, \alpha \approx 62°22'$$

$$\tan\alpha = \frac{4-\sqrt{3}}{3}, \alpha \approx 37°5'$$

15 With the Cartesian equation, we can derive the general equation for the range of a projectile. Using $y = -\frac{gx^2}{2V^2}\sec^2\theta + x\tan\theta$, show, using the properties of the sine graph, that the range of an object projected from ground level is $x = \frac{V^2}{g}\sin 2\theta$, and explain why the maximum range occurs at an angle of projection of $45°$.

a) $y = 0,\ x \neq 0$:

$$x\tan\theta = \frac{gx^2}{2V^2}\sec^2\theta$$

$$x = \frac{2V^2\tan\theta}{g\sec^2\theta}$$

$$= \frac{2V^2\sin\theta\cos\theta}{g}$$

$$= \frac{V^2\sin 2\theta}{g}$$

b) $-1 \leq \sin 2\theta \leq 1$

$\therefore$ max range $= \frac{V^2}{g}$

$\sin 2\theta = 1$ at $2\theta = 90°$

$\theta = 45°$

16 By rewriting $y = -\frac{gx^2}{2V^2}(1 + \tan^2\theta) + x\tan\theta$ as a quadratic in $\tan\theta$, show that a projectile can pass through the point $P(x, y)$ only if $y \leq \frac{V^4 - g^2x^2}{2V^2g}$ and explain what this means geometrically.

$$y = \frac{-gx^2}{2V^2} + \frac{-gx^2}{2V^2}\tan^2\theta + x\tan\theta$$

$$2V^2y = -gx^2 - gx^2\tan^2\theta + 2V^2x\tan\theta$$

$$gx^2\tan\theta - 2V^2x\tan\theta + 2V^2y + gx^2 = 0$$

For this equation to be true, there must be real values of θ that satisfy it.

In other words, $\Delta \geq 0$

$$(-2V^2x)^2 - 4(gx^2)(2V^2y + gx^2) \geq 0$$

$$4V^4x^2 - 8gx^2V^2y - 4g^2x^4 \geq 0$$

$$-8gx^2V^2y \geq -4V^4x^2 + 4g^2x^4$$

$$y \leq \frac{-4V^4x^2 + 4g^2x^4}{-8gx^2V^2} \quad \begin{matrix} \div -4x^2 \\ \div -4x^2 \end{matrix}$$

$$y \leq \frac{V^4 - g^2x^2}{2gV^2}$$

This means all points on or below the parabola $y = \frac{V^4 - g^2x^2}{2gV^2}$ can be reached by some variable launch angle θ.

Points above it cannot be reached because no real launch angle exists.

https://www.desmos.com/calculator/tzqwu4bxsz